Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Implementation Solution Guide

Version 1.0.0 · 2026-07-07

Contents

Executive Summary — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 5

What you get . . . 5

What this delivers . . . 5

The deployment model . . . 5

Outcome . . . 5

1. Solution Overview — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 6

What this solution is . . . 6

Why this matters . . . 6

What DSPM and DSPM for AI actually show . . . 6

The scope of "outcomes" (what this solution does and does not decide) . . . 7

Naming convention (used in every doc and script) . . . 7

The standard deployment model . . . 8

2. Requirements Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 8

2.1 DSPM / DSPM-for-AI licensing (verify in your tenant) . . . 8

2.2 DSPM roles (least-privilege) and portal access . . . 8

2.3 Purview auditing (unified audit log) turned on . . . 9

2.4 A labels + DLP foundation to build posture on . . . 9

2.5 Microsoft 365 Copilot / AI apps in scope (for DSPM for AI) . . . 9

2.6 SharePoint/OneDrive access & sharing signal (for oversharing) . . . 9

2.7 Pilot scope and simulation/test discipline . . . 10

Global Requirements — Microsoft Purview Solutions 10

1. Microsoft 365 tenant with the required Purview/compliance licensing . . . 10

2. Purview roles (least-privilege) and portal access . . . 11

3. Security & Compliance PowerShell + Graph toolchain . . . 11

4. Data sources & scope in scope . . . 11

5. Compliance, records, privacy & legal stakeholders . . . 11

6. A pilot scope and change control . . . 12

3. Implementation Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 12

Step 0 — Confirm scope, licensing, roles, and connections (Assess) . . . 12

Step 1 — Inventory the controls that feed posture (Assess) . . . 13

Step 2 — Read the DSPM posture and sensitive-data risk (Assess) . . . 13

Step 3 — Turn on / open DSPM for AI (Assess) . . . 13

Step 4 — Assess AI/Copilot data-access and sensitive interactions (Assess) . . . 14
Step 5 — Capture the recommendation/action baseline (Design) . . . 14
Step 6 — Act on recommendations and pilot oversharing reduction (Configure → Pilot) . . . 14
Step 7 — Validate the pilot (Validate) . . . 15
Step 8 — Expand and enforce (Expand) . . . 15
5. Testing Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 15
Test 1 — Environment, licensing, roles, and auditing . . . 16
Test 2 — Controls inventory is complete . . . 16
Test 3 — Posture and recommendations render . . . 16
Test 4 — DSPM for AI reports populate . . . 16
Test 5 — Oversharing targets are identifiable . . . 16
Test 6 — Recommendation baseline captured . . . 16
Test 7 — Reduction helper makes no change in dry-run . . . 16
Test 8 — Pilot policies are created in simulation/test only . . . 17
Test 9 — Simulation/test results are acceptable (enforcement gate) . . . 17
Test 10 — Recommendations move and oversharing drops on re-measure . . . 17
Test 11 — Rollback works . . . 17
6. Troubleshooting Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 17
DSPM or DSPM for AI blade is not visible . . . 17
`PURVDSPM-ScopeConfig.ps1` reports the S&C; session as FAIL . . . 17
DSPM for AI reports are empty . . . 18
`Search-UnifiedAuditLog` returns nothing or errors . . . 18
Auto-labeling or DLP cmdlet fails as "not recognized" / access denied . . . 18
`PURVDSPM-Reduce-Oversharing.ps1 -Action DlpForAI` stops asking for `-DlpAiLocation` . . . 18
A pilot policy landed in an enforcing state . . . 18
Simulation/test shows too many false positives . . . 19
Recommendations don't change after acting . . . 19
Enforcement broke legitimate collaboration . . . 19
Escalation boundary . . . 19
7. Best Practices — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 19
8. Deliverables Summary — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI) 20
Documentation . . . 21
PowerShell scripts . . . 21
What the buyer receives . . . 22

Outcome 22
Appendix A — PowerShell Scripts 23

Executive Summary — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

You cannot reduce data-security risk you cannot see — and generative AI has made that blind spot dangerous. Microsoft 365 Copilot can surface any content a user already has access to, so years of over-shared, unlabeled files that "nobody ever found" are suddenly one prompt away. This solution implements **Data Security Posture Management (DSPM)** in Microsoft Purview to make the risk visible and reduce it: the DSPM posture and recommendations, the sensitive-data risk across your estate, and **DSPM for AI** — visibility into what Copilot and other AI apps can access, which prompts and responses touch sensitive data, and the controls to reduce oversharing before and during AI adoption.

This is a complete, deploy-it-today package written for a competent Purview / compliance administrator, using only the Microsoft Purview portal, Security & Compliance PowerShell, and Microsoft Graph — no third-party tooling. Every high-impact change is piloted in **simulation/test** before it is enforced.

What you get

• A plain-language Solution Overview of what DSPM and DSPM for AI show, the availability/licensing caveat

for these evolving features, and the boundary between technical method and legal/records determination.

• A Requirements Guide with verify / implement-if-missing for every prerequisite — licensing, DSPM roles,

auditing, the labels + DLP foundation, Copilot/AI in scope, the sharing signal, and pilot discipline.

• An Implementation Guide of precise, ordered portal and PowerShell steps: read the posture, assess

sensitive-data and AI/Copilot exposure, act on recommendations, pilot each reduction in simulation/test, validate, expand to enforce, and review on a recurring cadence.

• Five PowerShell scripts for scope/health checking, controls discovery, AI/oversharing exposure

reporting, recommendation tracking, and safe oversharing reduction (read-only or `-WhatIf` by default).

• Testing and Troubleshooting guides, with an explicit enforcement gate, for a confident, safe rollout.

What this delivers

A measured posture baseline across your data-security controls, a sensitive-data risk assessment, a DSPM-for-AI exposure report of what Copilot/AI can reach and what it has touched, a prioritized set of oversharing targets, and a validated reduction workflow — auto-labeling in simulation, DLP for AI in test, and sharing restriction — that improves posture on re-measure rather than merely adding policies.

The deployment model

Assess → Design → Configure → Pilot (simulate/test) → Validate → Expand (enforce).

Outcome

Data-security risk made visible and reduced — posture measured, recommendations actioned, and Copilot/AI oversharing surfaced and controlled — so data is safer before and during AI adoption. Feature availability is flagged to verify in your tenant; regulatory, records, and legal determinations remain with your governance functions.

1. Solution Overview — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

What this solution is

You cannot reduce data-security risk you cannot see. **Data Security Posture Management (DSPM)** in Microsoft Purview is the place that *makes the risk visible* — it aggregates signals from your sensitivity labels, Data Loss Prevention (DLP), auto-labeling, and Insider Risk configuration into a single posture view, surfaces where **sensitive data** is exposed across the estate, and hands you **recommendations** to close the gaps. **DSPM for AI** extends the same idea to generative AI: it shows what Microsoft 365 Copilot and other AI apps can *access*, which **prompts and responses touch sensitive data**, and gives you the **oversharing-reduction controls** (sensitivity labels, DLP, and SharePoint access) to shrink that exposure before and during AI adoption.

This solution is a repeatable, self-guided way to **enable DSPM and DSPM for AI, read the posture, act on the recommendations, and reduce Copilot/AI oversharing** — using only the Microsoft Purview portal, Security & Compliance PowerShell (`Connect-IPPSSession`), and Microsoft Graph, with no third-party tooling. Every high-impact change is **piloted in simulation/test first**.

Why this matters

- **Risk is invisible by default.** Sensitive data is scattered across SharePoint, OneDrive, Teams, and Exchange, much of it unlabeled and over-shared. DSPM is the first view that quantifies it.
- **Generative AI changes the blast radius.** Copilot can surface any content a user already has access to. A file that was "technically shared with everyone but nobody ever found it" becomes one prompt away. DSPM for AI makes that exposure measurable *before* it becomes an incident.
- **Recommendations are only value once actioned.** DSPM proposes concrete fixes (turn on a label, extend DLP, restrict sharing). The worth is in working the recommendation queue, not admiring it.
- **Oversharing reduction is high-impact.** Broad-scope sharing changes, new DLP, and auto-labeling can disrupt real work if enforced blindly — so this solution pilots every reduction in simulation/test/disabled mode and validates before enforcing.

What DSPM and DSPM for AI actually show

- **DSPM (posture & recommendations)** — an at-a-glance posture across data-security controls, trends of

sensitive-data activity, and a prioritized **recommendations** list (e.g. label coverage gaps, DLP scope gaps, risky activity) with a status you work to completion.

- **Sensitive-data risk** — where labeled/unlabeled sensitive information lives and moves, so you can target labeling and DLP at the real exposure rather than guessing.
- **DSPM for AI — data access** — visibility into what Microsoft 365 Copilot and other AI apps can reach, including where broad/organization-wide access ("oversharing") makes sensitive content reachable by AI.
- **DSPM for AI — sensitive interactions** — reports of **prompts and responses** that reference sensitive data (sourced from Purview auditing), so you can see AI's actual contact with sensitive content, not just its theoretical reach.
- **DSPM for AI — oversharing controls** — the one-click/recommended policies to reduce exposure: sensitivity-label coverage, DLP for AI (keeping labeled content out of AI processing), and SharePoint site-access/sharing restrictions.

> ***Feature availability & licensing — verify in your tenant.*** *DSPM and DSPM for AI are newer and* ***actively evolving*** *capabilities. Which reports, recommendations, and one-click policies appear — and whether a capability is included or needs* ***Microsoft 365 E5 / the E5 Compliance add-on*** *(and in some cases* ***SharePoint Advanced Management*** *for site-access controls, or an AI/Copilot license) —* ***depends on your licensing and tenant****. Treat the specific blades and policy names in this guide as a current-state map to confirm against your own portal, never as guaranteed. This solution flags where to verify rather than asserting availability.*

The scope of "outcomes" (what this solution does and does not decide)

This solution delivers the **technical method and the evidence** — how to enable DSPM/DSPM for AI, read the posture, and configure the reduction controls. It does **not** make **regulatory, records, or legal determinations**. Which data must be retained or restricted, what an acceptable AI-exposure level is, and any records/legal-hold obligations are **deferred to your compliance, records, privacy, and legal functions**. You provide the configuration and the evidence; they own the determination.

Naming convention (used in every doc and script)

Every artifact this solution creates or exports is named `<Purpose> – <Scope> – <Stage>` and reused verbatim across the docs, scripts, and portal so a reader can trace one artifact end to end.

Artifact	Name
Posture & controls inventory export	`DSPM Posture – Estate – Discovery`
Sensitive-data risk assessment export	`Sensitive-Data Risk – Estate – Assess`
AI/Copilot data-access + prompts report	`AI Exposure – Copilot – Assess`
SharePoint/OneDrive oversharing export	`Oversharing Report – Sites – Assess`
Recommendation/action tracker export	`DSPM Recommendations – Estate – Tracker`
Pilot auto-labeling (simulation) policy	`Auto-Label Sensitive – Pilot – Simulation`

Artifact	Name
Pilot DLP-for-AI (test) policy	`DLP for AI – Pilot – Test`
Pilot oversharing-reduction scope	`Oversharing Reduction – Pilot – Stage1`
Broad oversharing-reduction scope	`Oversharing Reduction – Broad – Stage2`
Recurring posture-review cadence tag	`DSPM Posture Review – Estate – Recurring`

The standard deployment model

> *Enable and read the posture → assess sensitive-data and AI/Copilot exposure → act on recommendations → pilot every reduction in **simulation/test**, validate, then **expand to enforce** → review on a recurring cadence. Pattern: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce).***

2. Requirements Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Verify **every** requirement before configuring. Each entry gives **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites (tenant licensing, Purview roles, the S&C PowerShell + Graph toolchain, in-scope workloads, stakeholders, pilot/change-control) are in `global-requirements-purview.md` — this guide adds the DSPM/DSPM-for-AI-specific prerequisites.

> ***Availability caveat.** DSPM and DSPM for AI are newer, evolving capabilities whose blades, reports, and one-click policies change and are **licensing-/tenant-dependent**. Where this guide names a portal path or a policy, **confirm it exists in your tenant** rather than assuming — several requirements below are "verify in your tenant" by design.*

2.1 DSPM / DSPM-for-AI licensing (verify in your tenant)

Description: The compliance licensing DSPM and DSPM for AI require — commonly **Microsoft 365 E5** or the **E5 Compliance** add-on for the full posture, recommendations, and AI reporting; some AI-exposure and site-access controls may additionally require **SharePoint Advanced Management** and/or a **Microsoft 365 Copilot** license. **Why:** Most DSPM signals derive from advanced Purview features (auto-labeling, advanced DLP, Insider Risk) that are plan-gated; the AI reports depend on AI/Copilot being licensed and in use. **Verify:** In the Microsoft 365 admin center confirm the subscriptions/add-ons; in the Purview portal (`https://purview.microsoft.com`) confirm **Data Security Posture Management** and **DSPM for AI** are visible and their reports render. Run `PURVDSPM-ScopeConfig.ps1` (read-only) to enumerate the labels, DLP, and auto-labeling features that will feed posture. **Implement:** Assign/purchase the required licensing before starting; if a capability is absent, note it as out of scope and proceed with the licensed subset.

2.2 DSPM roles (least-privilege) and portal access

Description: Membership in the least-privilege Purview role group(s) that grant DSPM and DSPM-for-AI access — typically an information-protection / DSPM-capable role group for reading posture, plus the

matching configuration role for changing labels, DLP, or auto-labeling. Confirm the **exact current role-group names** in your portal. **Why:** Purview uses fine-grained role groups; reading posture, working recommendations, and creating policies each need the matching role, and separation of duties should be preserved. **Verify:** Purview portal → **Settings / Roles & scopes** → **Role groups** — confirm your account's membership; confirm you can open **Data Security Posture Management** and **DSPM for AI** without an access error. **Implement:** Assign the least-privilege role group under change control; reserve Global/Compliance Administrator for the initial enablement only.

2.3 Purview auditing (unified audit log) turned on

Description: Purview **Audit** (the unified audit log) enabled tenant-wide, since DSPM-for-AI prompt/response reporting and much of the activity signal is sourced from audit records. **Why:** Without auditing on, AI-interaction and sensitive-activity reporting is empty or incomplete. **Verify:** Purview portal → **Audit** shows searching is on; `PURVDSPM-ScopeConfig.ps1` reports whether auditing is enabled. AI-interaction audit records should be present once Copilot/AI is used. **Implement:** Turn on auditing (a one-time tenant action); allow time for records to accumulate before expecting AI reports to populate.

2.4 A labels + DLP foundation to build posture on

Description: At least a **baseline of sensitivity labels and DLP** in place (or ready to create), because DSPM posture and DSPM-for-AI reduction controls act *through* labels and DLP — DSPM measures and recommends; labels/DLP are the levers. **Why:** Recommendations such as "extend a label" or "protect sensitive data in Copilot" have no effect without a label taxonomy and DLP to apply; oversharing reduction relies on them. **Verify:** Purview portal → **Information Protection** → **Labels** and **Data Loss Prevention** → **Policies** show your taxonomy and policies; `PURVDSPM-Discovery.ps1` inventories labels, label policies, DLP policies, and auto-labeling policies. **Implement:** Stand up a baseline label taxonomy and at least one DLP policy first (a dedicated sensitivity/retention-labels or DLP solution covers this) so DSPM has levers to recommend.

2.5 Microsoft 365 Copilot / AI apps in scope (for DSPM for AI)

Description: Clarity on which AI is in scope — **Microsoft 365 Copilot**, Copilot experiences, and/or other **enterprise AI apps** — and whether it is licensed and in use, so DSPM-for-AI reporting has data. **Why:** DSPM for AI reports on the AI actually present; with no Copilot/AI usage there is nothing to report yet, and the value is pre-adoption posture plus in-adoption monitoring. **Verify:** Confirm Copilot/AI licensing and rollout status with the owning team; Purview portal → **DSPM for AI** shows connected AI activity once present. `PURVDSPM-AIExposure.ps1` reports what AI activity is visible. **Implement:** Confirm the AI in scope with stakeholders; if Copilot is not yet deployed, use DSPM for AI in its **pre-adoption** mode to reduce oversharing *before* turning Copilot on.

2.6 SharePoint/OneDrive access & sharing signal (for oversharing)

Description: The ability to read **site/file access and sharing** state — the "oversharing" that makes content reachable by AI — and, for the strongest site-access controls, **SharePoint Advanced Management** (verify licensing). **Why:** Oversharing reduction targets broadly-shared and organization-wide-accessible content; measuring and restricting it needs the sharing/access signal and the SharePoint controls. **Verify:** `PURVDSPM-AIExposure.ps1` reports SharePoint/OneDrive sharing exposure from the signals available to it; confirm whether SharePoint Advanced Management site-access

reviews/restrictions are available in your tenant. **Implement:** Where the advanced site-access controls are unavailable, use the label + DLP-for-AI levers plus manual site-owner sharing reviews as the reduction path.

2.7 Pilot scope and simulation/test discipline

Description: A named **pilot scope** (a ring of users/sites) and the intent to run every high-impact reduction — auto-labeling, DLP for AI, sharing restriction — in **simulation/test/disabled** mode first (as the category model requires, GR §6). **Why:** Auto-labeling and DLP can misfire on real content; a broad sharing restriction can break legitimate collaboration. Simulation/test surfaces the impact before anyone is affected. **Verify:** Confirm the pilot ring, that auto-labeling **simulation mode** and DLP **test mode** are available, and that a change window/approval exists. **Implement:** Define the pilot and the change-control path before enforcing anything; the scripts default to `-WhatIf` and create policies in simulation/test.

Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy, and legal functions** — the solutions provide the technical configuration method and evidence, not the determination. Automation is **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with the required Purview/compliance licensing

- **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

- **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

- **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

- **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Purview roles (least-privilege) and portal access

- **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

- **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.
- **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.
- **How to implement if missing:** Assign the least-privilege role group under change control.

3. Security & Compliance PowerShell + Graph toolchain

- **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

- **Why required:** Configuration, automation, and the provided scripts run through these.
- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams,

endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.

- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.
- **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with

stakeholders.

- **How to implement if missing:** Define the scope and classification intent before authoring policies.

5. Compliance, records, privacy & legal stakeholders

- **Description:** Engagement with the **compliance, records-management, privacy, security, and legal**

stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).

- **Why required:** Purview configuration must align to enterprise policy and legal/records requirements,

not technical defaults; several capabilities have HR/legal sensitivity.

- **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.
- **How to implement if missing:** Identify the owners and capture the requirements before configuring.

6. A pilot scope and change control

• **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate

on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.

• **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate

→ Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.

• **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change

window/approval path.

• **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant governance functions.*

3. Implementation Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Follow these steps **in order**. Posture and reports are read in the **Microsoft Purview portal** (`https://purview.microsoft.com`); inventory, exposure, and reduction are automated with Security & Compliance PowerShell (`Connect-IPPSSession`) and Microsoft Graph. Artifact names follow the `<Purpose> – <Scope> – <Stage>` convention from the Overview and are reused verbatim. **Every high-impact reduction is piloted in simulation/test first** — the scripts default to `-WhatIf` and to simulation/test mode.

> ***Model:*** *Enable and read the posture, assess sensitive-data and AI/Copilot exposure, act on recommendations, pilot each reduction in simulation/test, validate, then expand to enforce, and review on a recurring cadence. Pattern:* ***Assess → Design → Configure → Pilot (simulate/test) → Validate → Expand (enforce). Portal-path caveat.*** *The DSPM and DSPM-for-AI blades are evolving; the paths below are the current map. If a blade or policy name differs in your tenant, use the closest equivalent and* ***confirm before acting*** *— do not assume a path that is not present.*

Step 0 — Confirm scope, licensing, roles, and connections (Assess)

1. Run `PURVDSPM-ScopeConfig.ps1` (read-only). It confirms your `Connect-IPPSSession`/`Connect-MgGraph`

sign-in, reports whether **auditing** is on, and enumerates the licensing-dependent features (labels, DLP, auto-labeling, Insider Risk presence) that will feed posture — resolving Requirements 2.1–2.3.

1. In the Purview portal confirm **Data Security Posture Management** and **DSPM for AI** are visible

(Requirement 2.1). Note any capability that is absent so you scope around it.

1. Confirm the **pilot ring** and the change window (Requirement 2.7) before any configuration.

Pass/fail: Sign-in succeeds, auditing is on, DSPM and DSPM for AI are visible, pilot scope agreed. **Rollback:** None — read-only.

Step 1 — Inventory the controls that feed posture (Assess)

1. Run `PURVDSPM-Discovery.ps1` (read-only). It inventories **sensitivity labels**, **label policies**,

DLP policies/rules, and **auto-labeling policies** via S&C PowerShell and writes the export `DSPM Posture – Estate – Discovery`.

1. Review the export against Requirement 2.4 — this is your baseline of the levers DSPM will recommend

extending. A thin taxonomy or narrow DLP scope is itself the first finding.

Pass/fail: The export lists your labels, label policies, DLP policies, and auto-labeling policies. **Rollback:** None — read-only.

Step 2 — Read the DSPM posture and sensitive-data risk (Assess)

1. Purview portal → **Data Security Posture Management** → **Overview**. Read the posture summary and the

sensitive-data activity trends. This is the estate-level "where is the risk" view.

1. Open **Recommendations**. Each recommendation (label-coverage gap, DLP-scope gap, risky activity,

etc.) has a status — this is the queue you will work in Step 6.

1. Run `PURVDSPM-Discovery.ps1 -IncludeSensitiveTypes` to append the **sensitive information types**

in scope to `Sensitive-Data Risk – Estate – Assess`, giving a portable record of what "sensitive" means in your tenant to accompany the portal view.

Pass/fail: Posture overview renders; a recommendations list is present; the risk export is written. **Rollback:** None — read-only.

Step 3 — Turn on / open DSPM for AI (Assess)

1. Purview portal → **DSPM for AI** → **Overview**. If prompted, complete the one-time **get-started /

enablement** so AI reporting begins (this may enable audit collection for AI interactions — confirm it aligns with Requirement 2.3).

1. Confirm the AI in scope (Requirement 2.5) is reflected — Microsoft 365 Copilot and any other AI apps

Purview can see. If Copilot is not yet deployed, you are in **pre-adoption** posture mode: the goal is to reduce oversharing *before* enabling Copilot.

Pass/fail: DSPM for AI overview loads; in-scope AI is reflected (or pre-adoption mode confirmed). **Rollback:** Enablement is non-destructive; no policy is enforced by opening the blade.

Step 4 — Assess AI/Copilot data-access and sensitive interactions (Assess)

1. In **DSPM for AI** → **Reports**, review the AI **data-access / oversharing** view (what AI can reach) and

the **sensitive interactions** view (prompts/responses referencing sensitive data). These are the two exposure questions: *what can AI reach* and *what has AI actually touched*.

1. Run `PURVDSPM-AIExposure.ps1` (read-only). It reports the AI-interaction activity available from

Purview auditing and the **SharePoint/OneDrive oversharing** signal, writing `AI Exposure – Copilot – Assess` and `Oversharing Report – Sites – Assess`.

1. Cross-read the two: broadly-shared sites holding **unlabeled sensitive** content are the priority

oversharing targets, because Copilot can surface them to anyone with access.

Pass/fail: Both reports are written; you can name the top oversharing targets. **Rollback:** None — read-only.

Step 5 — Capture the recommendation/action baseline (Design)

1. Run `PURVDSPM-Tracker.ps1` (read-only) to snapshot the current DSPM recommendations and the DLP /

auto-labeling policy states into `DSPM Recommendations – Estate – Tracker` — the before-state you will drive to completion and re-measure against.

1. With compliance/records/privacy/legal stakeholders (GR §5), agree which recommendations to act on and

what an acceptable AI-exposure level is. **The determination is theirs**; you provide the evidence and will implement the agreed technical changes.

Pass/fail: Tracker baseline written; the agreed action set is recorded. **Rollback:** None — read-only.

Step 6 — Act on recommendations and pilot oversharing reduction (Configure → Pilot)

Work the agreed set. Every high-impact lever is piloted in **simulation/test** on the pilot ring first.

1. **Auto-labeling (simulation).** Run

`PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -Scope "Oversharing Reduction – Pilot – Stage1"` (defaults to `-WhatIf`; add `-Commit` to create it). With `-Commit` it creates the auto-labeling policy `Auto-Label Sensitive – Pilot – Simulation` in **simulation mode** — it labels *nothing* yet, it only shows what it *would* label. This closes label-coverage recommendations without risk.

1. **DLP for AI (test).** Run

`PURVDSPM-Reduce-Oversharing.ps1 -Action DlpForAI -Scope "Oversharing Reduction – Pilot – Stage1"` with `-Commit` to create the DLP policy `DLP for AI – Pilot – Test` in **test/disabled** mode, scoped to keep the agreed sensitivity-labeled content out of AI processing. **Verify the Copilot/AI DLP location name in your portal** — the script parameterizes it and does not assume it (see the script's notes).

1. **Sharing/site-access restriction.** Where the site-access controls are available (Requirement 2.6),

restrict the top oversharing targets from `Oversharing Report – Sites – Assess` — do this on the pilot sites first via site owners / the portal, and record the change. The script's `-Action SharingReview` produces the prioritized site list to act on; it does **not** silently change sharing.

Pass/fail: Pilot auto-label policy is in **simulation**, DLP for AI is in **test/disabled**, and the pilot sharing list is produced — nothing is enforced yet. **Rollback:** Delete the pilot policies (they are simulation/test and enforce nothing); no content was changed.

Step 7 — Validate the pilot (Validate)

1. Let simulation/test run the agreed window. In the portal review the auto-labeling **simulation

results **(what it would label) and the DLP** test** matches (what it would block/flag) for false positives.

1. Re-run `PURVDSPM-AIExposure.ps1` and `PURVDSPM-Tracker.ps1`; compare against

`DSPM Recommendations – Estate – Tracker` — the actioned recommendations should show progress and the pilot oversharing signal should be lower.

1. Confirm with stakeholders that the simulated outcomes match intent (no legitimate collaboration

broken). See Testing (doc 5) for the explicit gates.

Pass/fail: Simulation/test results reviewed, acceptable false-positive rate, recommendations show movement. **Rollback:** If results are wrong, adjust the policy conditions and re-simulate — enforce nothing until the gate passes.

Step 8 — Expand and enforce (Expand)

1. Only after the pilot gate passes: turn the auto-labeling policy from **simulation** to **on**, and the

DLP-for-AI policy from **test** to **enforce**, and widen scope to `Oversharing Reduction – Broad – Stage2`. Do this as a deliberate, change-controlled step — `PURVDSPM-Reduce-Oversharing.ps1 -Action Enforce -Commit` performs the mode change on the named policies (still gated behind `-Commit`).

1. Apply the broad sharing/site-access restrictions to the remaining oversharing targets under change

control.

1. Stand up the recurring review: tag the cadence `DSPM Posture Review – Estate – Recurring` and schedule

the read-only `ScopeConfig → Discovery → AIExposure → Tracker` chain (e.g. monthly) so posture stays measured as data and AI usage change.

Pass/fail: Policies enforced at broad scope, oversharing reduced on re-measure, recurring review scheduled. **Rollback:** Return the enforced policy to simulation/test (or delete it) and restore prior sharing on any site where enforcement caused an issue; the tracker export is the before-state to restore toward.

5. Testing Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Run these after each stage. Every test is safe (read-only or simulation/test); the simulation and test reviews are the **gates** before any enforcement. Pass/fail is explicit.

Test 1 — Environment, licensing, roles, and auditing

Do: Run `PURVDSPM-ScopeConfig.ps1`; open **DSPM** and **DSPM for AI** in the Purview portal. **Expected:** Sign-in succeeds; auditing reports as **on**; both DSPM blades load without an access error; the licensing-dependent features are enumerated. Any absent capability is noted, not assumed. **Gate:** Do not proceed until DSPM and DSPM for AI are visible and auditing is on.

Test 2 — Controls inventory is complete

Do: Run `PURVDSPM-Discovery.ps1`; compare against **Information Protection** → **Labels** and **DLP** → **Policies** in the portal. **Expected:** `DSPM Posture – Estate – Discovery` lists every sensitivity label, label policy, DLP policy, and auto-labeling policy shown in the portal — the baseline of levers DSPM will recommend.

Test 3 — Posture and recommendations render

Do: Open **DSPM** → **Overview** and **Recommendations**; run `PURVDSPM-Discovery.ps1 -IncludeSensitiveTypes`. **Expected:** The posture overview and sensitive-data trends display; a recommendations list is present; `Sensitive-Data Risk – Estate – Assess` records the in-scope sensitive information types.

Test 4 — DSPM for AI reports populate

Do: Open **DSPM for AI** → **Reports**; run `PURVDSPM-AIExposure.ps1`. **Expected:** The AI data-access and sensitive-interaction views render (or, pre-adoption, show the oversharing exposure with no AI activity yet); `AI Exposure – Copilot – Assess` and `Oversharing Report – Sites – Assess` are written. *Note:* AI-interaction reporting depends on auditing being on (Test 1) and on Copilot/AI usage having occurred.

Test 5 — Oversharing targets are identifiable

Do: Cross-read `Oversharing Report – Sites – Assess` with the AI data-access view. **Expected:** You can name the top oversharing targets — broadly-shared sites holding unlabeled sensitive content — as the priority reduction set.

Test 6 — Recommendation baseline captured

Do: Run `PURVDSPM-Tracker.ps1`. **Expected:** `DSPM Recommendations – Estate – Tracker` snapshots the current recommendations and the DLP / auto-labeling policy states — the before-state to re-measure against.

Test 7 — Reduction helper makes no change in dry-run

Do: Run `PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -WhatIf` and `... -Action DlpForAI -WhatIf`; then re-run `PURVDSPM-Discovery.ps1`. **Expected:** The dry-run prints the

intended policy it *would* create; the discovery inventory is unchanged — `-WhatIf` created nothing. **This is the gate that proves the helper is safe by default.**

Test 8 — Pilot policies are created in simulation/test only

Do: Run the helper with `-Commit` for the pilot (`Auto-Label Sensitive – Pilot – Simulation`, `DLP for AI – Pilot – Test`); inspect them in the portal. **Expected:** The auto-labeling policy is in **simulation** mode (labels nothing) and the DLP policy is in **test/disabled** mode (enforces nothing). Neither changes content. **Gate:** If either landed in an enforcing state, stop and correct before continuing.

Test 9 — Simulation/test results are acceptable (enforcement gate)

Do: Let the pilot run the agreed window; review auto-labeling **simulation results** and DLP **test** matches in the portal for false positives; confirm with stakeholders. **Expected:** Simulated labels/blocks match intent with an acceptable false-positive rate and no legitimate collaboration would be broken. **Gate: Do not enforce (Step 8 of implementation) until this passes.**

Test 10 — Recommendations move and oversharing drops on re-measure

Do: After acting on the agreed set, re-run `PURVDSPM-Tracker.ps1` and `PURVDSPM-AIExposure.ps1`; compare to the baseline. **Expected:** Actioned recommendations show progress/closed status and the pilot oversharing signal is lower — evidence the posture actually improved, not just that a policy exists.

Test 11 — Rollback works

Do: On a pilot policy, exercise the documented rollback (return to simulation/test or delete). **Expected:** The policy reverts with no residual content change; the tracker export confirms the before-state is restorable.

6. Troubleshooting Guide — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Field issues and fixes. Most stem from **licensing/availability** (these features are evolving and plan-gated), **auditing being off**, or the AI simply **not having been used yet**. When a portal path differs from this guide, confirm the current equivalent rather than assuming.

DSPM or DSPM for AI blade is not visible

- **Likely cause:** Licensing (Requirement 2.1) or role membership (Requirement 2.2).
- **Fix:** Confirm the subscription/add-ons in the M365 admin center; confirm your account is in a

DSPM-capable role group in Purview → Roles & scopes. Run `PURVDSPM-ScopeConfig.ps1` to enumerate what your session can see. If the capability is genuinely absent for your plan, scope around it and note it.

`PURVDSPM-ScopeConfig.ps1` reports the S&C session as FAIL

- **Cause:** Not connected to Security & Compliance PowerShell.
- **Fix:** `Connect-IPPSSession` with a modern-auth/MFA admin (GR §3). Re-run. For the auditing check and

`Search-UnifiedAuditLog`, also connect Exchange Online (`Connect-ExchangeOnline`).

DSPM for AI reports are empty

- **Cause 1 — auditing off:** Prompt/response and AI-interaction reporting is sourced from the unified

audit log. **Fix:** Turn auditing on (Requirement 2.3); `PURVDSPM-ScopeConfig.ps1` shows `UnifiedAuditLogIngestionEnabled`. Allow time for records to accumulate.

- **Cause 2 — no AI usage yet:** With no Copilot/AI activity there is nothing to report. **Fix:** This is

expected pre-adoption — use the oversharing signal (`PURVDSPM-AIExposure.ps1` sharing events) to reduce exposure before enabling Copilot.

- **Cause 3 — wrong audit RecordType/Operations:** The record types/operations that surface AI

interactions evolve. **Fix:** Confirm the current values in Purview → Audit and pass them via `-AiRecordTypes` / `-SharingOperations`.

`Search-UnifiedAuditLog` returns nothing or errors

- **Cause:** Not connected to Exchange Online, auditing off, or the window predates auditing being on.
- **Fix:** `Connect-ExchangeOnline`; confirm auditing on; widen `-Days` within the retained window. Note

audit data has ingestion latency — very recent activity may not appear yet.

Auto-labeling or DLP cmdlet fails as "not recognized" / access denied

- **Cause:** Auto-labeling / advanced DLP is unlicensed, or the role lacks configuration rights.
- **Fix:** Confirm licensing (Requirement 2.1) and the configuration role (Requirement 2.2). Reading is

possible with a lower role than creating; `PURVDSPM-Discovery.ps1` will warn where a feature isn't readable.

`PURVDSPM-Reduce-Oversharing.ps1 -Action DlpForAI` stops asking for `-DlpAiLocation`

- **This is intentional.** The DLP location that scopes a policy to Microsoft 365 Copilot / AI is

licensing-dependent and evolving, so the script will not assume it. **Fix:** Confirm the exact location name in Purview → DLP (or use the DSPM-for-AI one-click "protect sensitive data in Copilot" policy in the portal if present), then pass it via `-DlpAiLocation`.

A pilot policy landed in an enforcing state

- **Cause:** Wrong mode or a manual portal change.

- **Fix:** `PURVDSPM-Reduce-Oversharing.ps1` creates auto-labeling in **simulation**

(`TestWithoutNotifications`) and DLP in **test** by design. If a policy is enforcing before the gate, return it to test/simulation (Set-*Policy -Mode TestWithoutNotifications) and re-validate. Never enforce before Testing Test 9 passes.

Simulation/test shows too many false positives

- **Cause:** Conditions too broad (sensitive-info-type confidence, label scope).
- **Fix:** Tighten the auto-labeling rule conditions / DLP rule and **re-simulate** — do not enforce.

The whole point of simulation/test is to tune before anyone is affected.

Recommendations don't change after acting

- **Cause:** Posture re-evaluation latency, or the action didn't address the specific recommendation.
- **Fix:** Allow the portal time to re-evaluate; re-run `PURVDSPM-Tracker.ps1` and compare policy state to

the baseline. Confirm the change you made maps to the recommendation you intended to close (record the mapping in the tracker's Notes column).

Enforcement broke legitimate collaboration

- **Cause:** A sharing restriction or DLP-for-AI block was too broad.
- **Fix:** Execute the documented rollback — return the policy to test/simulation (or delete it) and

restore the prior sharing on the affected site. Use `DSPM Recommendations – Estate – Tracker` as the before-state, tune conditions, re-simulate, then re-expand.

Escalation boundary

Regulatory, records-retention, and legal questions — what *must* be restricted or retained, what AI-exposure level is acceptable, legal-hold implications — are **not** resolved here. Capture the technical evidence (the exports and tracker) and route the determination to your compliance, records, privacy, and legal functions (GR §5).

7. Best Practices — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

- **Reduce oversharing *before* you turn on Copilot.** The safest AI adoption is one where the data was

already tidied. Use DSPM for AI's pre-adoption posture to shrink broad access first — a file Copilot can't reach can't be surfaced by a prompt.

- **Always pilot in simulation/test.** Auto-labeling runs in **simulation** and DLP runs in **test**

precisely so you can see impact before anyone is affected. `-WhatIf` is the default on the reduction helper for the same reason — read every intended action before you `-Commit`, and enforce only after the gate passes.

- **Work the recommendation queue; don't admire the dashboard.** DSPM's value is realized only when

recommendations move to done. Snapshot the queue, agree the action set with stakeholders, act, then re-measure — the trend line, not the screenshot, is the outcome.

- **Fix the biggest exposure first.** Prioritize broadly-shared sites holding **unlabeled sensitive**

content — that intersection is where AI exposure is highest. Start there, not with the easiest tickets.

- **Let compliance/records/privacy/legal own the determination.** You provide the technical method and

the evidence; *what* must be restricted or retained and *what* AI-exposure level is acceptable are their calls. Capture the agreed set before configuring.

- **Verify availability, never assume it.** DSPM and DSPM for AI are evolving and licensing-dependent.

When a blade, report, or one-click policy isn't where this guide says, confirm it in your tenant and use the closest current equivalent — don't force a path that isn't there.

- **Keep auditing on.** DSPM-for-AI prompt/response reporting is only as complete as the unified audit

log feeding it. If AI reports look empty, check auditing and confirm AI usage has actually occurred.

- **Confirm the Copilot/AI DLP location in your portal.** The DLP-for-AI location name evolves; the

reduction helper parameterizes it rather than assuming it. Verify the exact location before you enforce a DLP-for-AI policy so it scopes where you intend.

- **Labels and DLP are the levers — keep them healthy.** DSPM measures and recommends, but acts through

your label taxonomy and DLP. A thin taxonomy caps how much posture you can improve; invest there.

- **Preserve separation of duties.** Read posture with a least-privilege role; keep configuration rights

separate. Reserve Global/Compliance Administrator for the initial enablement only.

- **Make posture review recurring.** Data and AI usage drift. Schedule the read-only `ScopeConfig → Discovery → AIExposure → Tracker chain under DSPM Posture Review – Estate – Recurring` so the estate stays measured, not measured once.

- **Keep an easy rollback.** Every enforced policy should be one change from returning to simulation/test.

Keep the tracker export as the before-state so any enforcement can be safely reverted.

8. Deliverables Summary — Microsoft Purview Data Security Posture Management (DSPM & DSPM for AI)

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What DSPM and DSPM for AI show (posture, sensitive-data risk, AI/Copilot data access, sensitive prompts/responses, oversharing controls), the availability/licensing caveat, the outcome-vs-determination boundary, and the naming convention
2. Requirements Guide	Every DSPM/DSPM-for-AI prerequisite with Verify / Implement-if-missing (licensing, DSPM roles, auditing, labels+DLP foundation, Copilot/AI in scope, sharing signal, pilot discipline)
Global Requirements (Purview)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered portal + PowerShell steps: confirm scope, inventory controls, read posture & risk, open DSPM for AI, assess AI exposure, baseline recommendations, pilot reduction in simulation/test, validate, expand/enforce, recurring review
5. Testing Guide	Validation procedures with explicit pass/fail and the enforcement gate
6. Troubleshooting Guide	Field issues and remediation
7. Best Practices	Field-proven recommendations for measuring and reducing posture safely

PowerShell scripts

All are safe-by-default (read-only, or `-WhatIf` / simulation-test unless `-Commit`), use `Connect-IPPSSession` and/or `Connect-MgGraph`, and store **no** credentials.

Script	What it does
`PURVDSPM-ScopeConfig.ps1`	Read-only scope/health helper: confirms sign-in, reports whether **auditing** is on, and enumerates the licensing-dependent features (labels, DLP, auto-labeling, Insider Risk presence) feeding posture. Resolves Requirements 2.1–2.3
`PURVDSPM-Discovery.ps1`	Read-only inventory of sensitivity labels, label policies, DLP policies/rules, and auto-labeling policies (and, with `-IncludeSensitiveTypes`, the sensitive info types). Writes `DSPM Posture – Estate – Discovery` and `Sensitive-Data Risk – Estate – Assess`

Script	What it does
`PURVDSPM-AIExposure.ps1`	Read-only DSPM-for-AI exposure report: AI/Copilot interaction activity from Purview auditing plus SharePoint/OneDrive oversharing signal. Writes `AI Exposure – Copilot – Assess` and `Oversharing Report – Sites – Assess`
`PURVDSPM-Tracker.ps1`	Read-only recommendation/action tracker: snapshots DSPM recommendations and DLP / auto-labeling policy states. Writes `DSPM Recommendations – Estate – Tracker`
`PURVDSPM-Reduce-Oversharing.ps1`	Oversharing-reduction helper (label / DLP-for-AI / sharing-review / enforce). Defaults to `-WhatIf`; with `-Commit` creates auto-labeling in **simulation** and DLP for AI in **test**, produces the prioritized sharing list, and (Enforce) flips named policies from simulation/test to on — never silently

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the model: **Assess → Design → Configure → Pilot (simulate/test) → Validate → Expand (enforce).**

Outcome

Data-security risk made **visible and reduced**: posture measured across the estate, recommendations actioned to completion, and Copilot/AI oversharing surfaced and controlled through labels, DLP for AI, and sharing restriction — every high-impact change piloted in simulation/test before enforcement — so data is safer before and during AI adoption. Regulatory, records, and legal determinations remain with the organization's governance functions; this solution delivers the technical method and the evidence.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

PURVDSPM-AIExposure.ps1

Purpose: The DSPM-for-AI exposure report. Pulls AI/Copilot interaction records from Purview auditing (the same signal that feeds DSPM for AI) and the SharePoint/OneDrive sharing signal that indicates oversharing — content broadly reachable, and therefore reachable by AI. Read-only. HOW TO RUN PowerShell 5.1+ or 7+ with Exchange Online / Security & Compliance PowerShell (for Search-UnifiedAuditLog): Connect-ExchangeOnline # (or Connect-IPPSSession where Search-UnifiedAuditLog is available) .\PURVDSPM-AIExposure.ps1 -Days 30 EXPECTED RESULTS AI-Exposure-Copilot-Assess.csv (AI interaction activity) and Oversharing-Report-Sites-Assess.csv (sharing-link / broad-access events). Empty AI results usually mean auditing is off (Req 2.3) or Copilot/AI has not been used yet — confirm both. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. VERIFY IN YOUR TENANT: the audit RecordType/Operations used to surface AI interactions and sharing evolve. This script parameterizes them (-AiRecordTypes / -SharingOperations) with current-documented defaults; confirm the exact values in Purview → Audit for your tenant rather than assuming. For the richest AI data-access / oversharing view, use the DSPM for AI reports and (where licensed) SharePoint Advanced Management's data-access governance report in the portal — this script provides the portable, scriptable signal, not a replacement for those blades. > [CmdletBinding()] param([int]$Days = 30, [string[]]$AiRecordTypes = @("CopilotInteraction"), [string[]]$SharingOperations = @("SharingSet","SharingInvitationCreated","AnonymousLinkCreated","AddedToSecureLink","CompanyLinkCreated"), [int]$ResultSize = 5000, [string]$AiOutPath = ".\AI-Exposure-Copilot-Assess.csv", [string]$SharingOutPath = ".\Oversharing-Report-Sites-Assess.csv")

How to run: PowerShell 5.1+ or 7+ with Exchange Online / Security & Compliance PowerShell (for Search-UnifiedAuditLog): Connect-ExchangeOnline # (or Connect-IPPSSession where Search-UnifiedAuditLog is available) .\PURVDSPM-AIExposure.ps1 -Days 30 EXPECTED RESULTS AI-Exposure-Copilot-Assess.csv (AI interaction activity) and Oversharing-Report-Sites-Assess.csv (sharing-link / broad-access events). Empty AI results usually mean auditing is off (Req 2.3) or Copilot/AI has not been used yet — confirm both. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. VERIFY IN YOUR TENANT: the audit RecordType/Operations used to surface AI interactions and sharing evolve. This script parameterizes them (-AiRecordTypes / -SharingOperations) with current-documented defaults; confirm the exact values in Purview → Audit for your tenant rather than assuming. For the richest AI data-access / oversharing view, use the DSPM for AI reports and (where licensed) SharePoint Advanced Management's data-access governance report in the portal — this script provides the portable, scriptable signal, not a replacement for those blades. > [CmdletBinding()] param([int]$Days = 30, [string[]]$AiRecordTypes = @("CopilotInteraction"), [string[]]$SharingOperations = @("SharingSet","SharingInvitationCreated","AnonymousLinkCreated","AddedToSecureLink","CompanyLinkCreated"), [int]$ResultSize = 5000, [string]$AiOutPath = ".\AI-Exposure-Copilot-Assess.csv", [string]$SharingOutPath = ".\Oversharing-Report-Sites-Assess.csv")

PURVDSPM-Discovery.ps1

Purpose: Reads the sensitivity labels, label policies, DLP policies and rules, and auto-labeling policies that Data Security Posture Management aggregates into posture — your baseline of the levers DSPM will recommend extending. With -IncludeSensitiveTypes it also records the sensitive information types in

scope. Read-only; changes nothing. HOW TO RUN PowerShell 5.1+ or 7+ with Security & Compliance PowerShell: Connect-IPPSSession .\PURVDSPM-Discovery.ps1 .\PURVDSPM-Discovery.ps1 -IncludeSensitiveTypes EXPECTED RESULTS DSPM-Posture-Estate-Discovery.csv listing labels, label policies, DLP policies/rules, and auto-labeling policies. With -IncludeSensitiveTypes, Sensitive-Data-Risk-Estate-Assess.csv of the sensitive info types. Compare counts against the Purview portal (Testing doc, Test 2). .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. Cmdlets are standard Security & Compliance PowerShell (Get-Label, Get-LabelPolicy, Get-DlpCompliancePolicy, Get-DlpComplianceRule, Get-AutoSensitivityLabelPolicy, Get-DlpSensitiveInformationType); none are invented. > [CmdletBinding()] param([switch]$IncludeSensitiveTypes, [string]$OutPath = ".\DSPM-Posture-Estate-Discovery.csv", [string]$SensitiveOutPath = ".\Sensitive-Data-Risk-Estate-Assess.csv")

How to run: PowerShell 5.1+ or 7+ with Security & Compliance PowerShell: Connect-IPPSSession .\PURVDSPM-Discovery.ps1 .\PURVDSPM-Discovery.ps1 -IncludeSensitiveTypes EXPECTED RESULTS DSPM-Posture-Estate-Discovery.csv listing labels, label policies, DLP policies/rules, and auto-labeling policies. With -IncludeSensitiveTypes, Sensitive-Data-Risk-Estate-Assess.csv of the sensitive info types. Compare counts against the Purview portal (Testing doc, Test 2). .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. Cmdlets are standard Security & Compliance PowerShell (Get-Label, Get-LabelPolicy, Get-DlpCompliancePolicy, Get-DlpComplianceRule, Get-AutoSensitivityLabelPolicy, Get-DlpSensitiveInformationType); none are invented. > [CmdletBinding()] param([switch]$IncludeSensitiveTypes, [string]$OutPath = ".\DSPM-Posture-Estate-Discovery.csv", [string]$SensitiveOutPath = ".\Sensitive-Data-Risk-Estate-Assess.csv")

PURVDSPM-Reduce-Oversharing.ps1

Purpose: The reduction helper for DSPM for AI. Depending on -Action it: creates an AUTO-LABELING policy in SIMULATION mode (labels nothing yet), creates a DLP policy for AI in TEST mode (enforces nothing), produces the prioritized SharingReview list, or flips named pilot policies from simulation/test to enforced. SAFE BY DEFAULT: prints intended actions under -WhatIf; makes changes only with -Commit, and even then creates policies in simulation/test — never enforcing on creation. HOW TO RUN PowerShell 5.1+ or 7+ with Security & Compliance PowerShell (Connect-IPPSSession). Dry-run (default, no change): .\PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -Scope "Oversharing Reduction - Pilot - Stage1" -SensitivityLabel "Confidential" -WhatIf Create pilot auto-labeling in SIMULATION: .\PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -SensitivityLabel "Confidential" -Commit Create pilot DLP for AI in TEST (VERIFY -DlpAiLocation for your tenant): .\PURVDSPM-Reduce-Oversharing.ps1 -Action DlpForAI -SensitivityLabel "Confidential" -DlpAiLocation "<verify-in-portal>" -Commit Produce the sharing-review target list from the oversharing export: .\PURVDSPM-Reduce-Oversharing.ps1 -Action SharingReview -OversharingCsv .\Oversharing-Report-Sites-Assess.csv Enforce (flip named pilot policies) AFTER the gate passes: .\PURVDSPM-Reduce-Oversharing.ps1 -Action Enforce -Commit EXPECTED RESULTS AutoLabel/DlpForAI: named policy exists in simulation/test (confirm in portal, Testing Test 8). SharingReview: Oversharing-Reduction-Targets.csv of prioritized sites. Enforce: named policies moved to on/enforce. Nothing is enforced without -Commit AND a deliberate -Action Enforce. .NOTES White-label Microsoft Purview tooling. SAFE-by-default (-WhatIf). No credentials stored. HIGH-IMPACT changes are piloted in simulation/test first per the category model — DO NOT enforce until the pilot gate (Testing Test 9) passes. VERIFY IN YOUR TENANT: the DLP location that scopes a policy to Microsoft 365 Copilot / AI evolves and is licensing-dependent — pass it via -DlpAiLocation after confirming the exact name in Purview → DLP; this script does NOT assume it. This helper does not change SharePoint sharing directly (SharingReview only produces the target list); apply site-access/sharing restrictions via site owners / the

portal / SharePoint Advanced Management under change control. > [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)] [ValidateSet("AutoLabel","DlpForAI","SharingReview","Enforce")] [string]$Action,

How to run: PowerShell 5.1+ or 7+ with Security & Compliance PowerShell (Connect-IPPSSession). Dry-run (default, no change): .\PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -Scope "Oversharing Reduction - Pilot - Stage1" -SensitivityLabel "Confidential" -WhatIf Create pilot auto-labeling in SIMULATION: .\PURVDSPM-Reduce-Oversharing.ps1 -Action AutoLabel -SensitivityLabel "Confidential" -Commit Create pilot DLP for AI in TEST (VERIFY -DlpAiLocation for your tenant): .\PURVDSPM-Reduce-Oversharing.ps1 -Action DlpForAI -SensitivityLabel "Confidential" -DlpAiLocation "<verify-in-portal>" -Commit Produce the sharing-review target list from the oversharing export: .\PURVDSPM-Reduce-Oversharing.ps1 -Action SharingReview -OversharingCsv .\Oversharing-Report-Sites-Assess.csv Enforce (flip named pilot policies) AFTER the gate passes: .\PURVDSPM-Reduce-Oversharing.ps1 -Action Enforce -Commit EXPECTED RESULTS AutoLabel/DlpForAI: named policy exists in simulation/test (confirm in portal, Testing Test 8). SharingReview: Oversharing-Reduction-Targets.csv of prioritized sites. Enforce: named policies moved to on/enforce. Nothing is enforced without -Commit AND a deliberate -Action Enforce. .NOTES White-label Microsoft Purview tooling. SAFE-by-default (-WhatIf). No credentials stored. HIGH-IMPACT changes are piloted in simulation/test first per the category model — DO NOT enforce until the pilot gate (Testing Test 9) passes. VERIFY IN YOUR TENANT: the DLP location that scopes a policy to Microsoft 365 Copilot / AI evolves and is licensing-dependent — pass it via -DlpAiLocation after confirming the exact name in Purview → DLP; this script does NOT assume it. This helper does not change SharePoint sharing directly (SharingReview only produces the target list); apply site-access/sharing restrictions via site owners / the portal / SharePoint Advanced Management under change control. > [CmdletBinding(SupportsShouldProcess = $true, ConfirmImpact = 'High')] param([Parameter(Mandatory = $true)] [ValidateSet("AutoLabel","DlpForAI","SharingReview","Enforce")] [string]$Action,

PURVDSPM-ScopeConfig.ps1

Purpose: Read-only pre-flight for Data Security Posture Management. Confirms your Security & Compliance PowerShell (and optional Graph) sign-in, reports whether unified auditing is on, and enumerates the licensing-dependent controls (sensitivity labels, DLP, auto-labeling, Insider Risk presence) that DSPM aggregates into posture. Changes nothing. HOW TO RUN PowerShell 5.1+ or 7+, with the ExchangeOnlineManagement (Security & Compliance PowerShell) and, optionally, Microsoft.Graph modules: Connect-IPPSSession .\PURVDSPM-ScopeConfig.ps1 To also probe Graph/auditing via Graph, pass -UseGraph (Connect-MgGraph is invoked read-only). EXPECTED RESULTS A readiness summary: sign-in OK, auditing on/off, and counts of labels / DLP / auto-labeling / Insider Risk artifacts present. Written to .\DSPM-ScopeConfig-Estate-Assess.csv. Resolves Requirements 2.1-2.3. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials are stored or requested in code. Availability caveat: DSPM / DSPM for AI blades and feature availability are licensing-/tenant- dependent and evolving — this script reports the controls it can see; confirm DSPM visibility in the Purview portal. Cmdlets used are standard Security & Compliance PowerShell; none are invented. > [CmdletBinding()] param([switch]$UseGraph, [string]$OutPath = ".\DSPM-ScopeConfig-Estate-Assess.csv")

How to run: PowerShell 5.1+ or 7+, with the ExchangeOnlineManagement (Security & Compliance PowerShell) and, optionally, Microsoft.Graph modules: Connect-IPPSSession .\PURVDSPM-ScopeConfig.ps1 To also probe Graph/auditing via Graph, pass -UseGraph (Connect-MgGraph is invoked read-only). EXPECTED RESULTS A readiness summary: sign-in OK, auditing on/off, and counts of labels / DLP / auto-labeling / Insider Risk artifacts present. Written to

.\DSPM-ScopeConfig-Estate-Assess.csv. Resolves Requirements 2.1-2.3. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials are stored or requested in code. Availability caveat: DSPM / DSPM for AI blades and feature availability are licensing-/tenant- dependent and evolving — this script reports the controls it can see; confirm DSPM visibility in the Purview portal. Cmdlets used are standard Security & Compliance PowerShell; none are invented. > [CmdletBinding()] param([switch]$UseGraph, [string]$OutPath = ".\DSPM-ScopeConfig-Estate-Assess.csv")

PURVDSPM-Tracker.ps1

Purpose: Captures the before/after state you drive to completion: the current DLP and auto-labeling policy states (the levers DSPM recommends) and their mode, so re-running after acting on recommendations shows movement. Because DSPM recommendation objects are surfaced primarily in the portal and via evolving APIs, this tracker records the policy-state evidence and lets you attach the portal recommendation status alongside. Read-only. HOW TO RUN PowerShell 5.1+ or 7+ with Security & Compliance PowerShell: Connect-IPPSSession .\PURVDSPM-Tracker.ps1 Re-run after acting on recommendations and compare to the prior snapshot (Testing doc, Test 10). EXPECTED RESULTS DSPM-Recommendations-Estate-Tracker.csv: each DLP / auto-labeling policy with its mode/enabled state and a Snapshot timestamp — the baseline to re-measure posture improvement against. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. VERIFY IN YOUR TENANT: DSPM's own recommendation list lives in Purview → DSPM → Recommendations; its programmatic surface is evolving and not assumed here. This tracker records the authoritative policy-state evidence via standard cmdlets (Get-DlpCompliancePolicy, Get-AutoSensitivityLabelPolicy); add the portal recommendation status in the Notes column as you work the queue. > [CmdletBinding()] param([string]$OutPath = ".\DSPM-Recommendations-Estate-Tracker.csv")

How to run: PowerShell 5.1+ or 7+ with Security & Compliance PowerShell: Connect-IPPSSession .\PURVDSPM-Tracker.ps1 Re-run after acting on recommendations and compare to the prior snapshot (Testing doc, Test 10). EXPECTED RESULTS DSPM-Recommendations-Estate-Tracker.csv: each DLP / auto-labeling policy with its mode/enabled state and a Snapshot timestamp — the baseline to re-measure posture improvement against. .NOTES White-label Microsoft Purview tooling. Read-only. No credentials stored. VERIFY IN YOUR TENANT: DSPM's own recommendation list lives in Purview → DSPM → Recommendations; its programmatic surface is evolving and not assumed here. This tracker records the authoritative policy-state evidence via standard cmdlets (Get-DlpCompliancePolicy, Get-AutoSensitivityLabelPolicy); add the portal recommendation status in the Notes column as you work the queue. > [CmdletBinding()] param([string]$OutPath = ".\DSPM-Recommendations-Estate-Tracker.csv")